

3.1 An Introduction to AS Level Organic Chemistry

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.1 An Introduction to AS Level Organic Chemistry
Difficulty	Medium

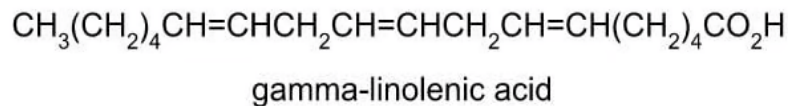
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Evening primrose oil has been shown to reduce high blood pressure and improve heart health. The oil is made from seeds, within which the active compound gamma-linolenic acid is found.



How many stereoisomers does gamma-linolenic acid have?

- A. 3
- B. 6
- C. 8
- D. 4

[1 mark]

Question 2

How many structural isomers are there of tribromopropane, $\text{C}_3\text{H}_5\text{Br}_3$?

- A. 4
- B. 5
- C. 6
- D. 7

[1 mark]

Question 3

Compound A, an organic compound with a molecular formula $\text{C}_6\text{H}_{12}\text{O}_2$, is produced by the oxidation of compound B, molecular formula $\text{C}_6\text{H}_{14}\text{O}$.

Which of the following functional groups does B contain?

- A. Carboxyl ($-\text{COOH}$)
- B. Carbonyl ($\text{C}=\text{O}$)
- C. Hydroxyl ($-\text{OH}$)
- D. Alkyl (RCOOR')

[1 mark]

Question 4

How many isomeric esters have the molecular formula $C_5H_{10}O_2$?

- A. 4
- B. 5
- C. 9
- D. 7

[1 mark]

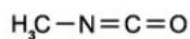
Question 5

Methyl isocyanate, CH_3NCO , is a toxic liquid which is used in the production of adhesives.

The sequence of atoms in a methyl isocyanate molecule is $H_3C-N=C=O$.

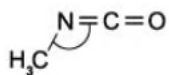
What is the approximate angle between the bonds formed by the N atom?

A.



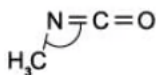
180°

B.



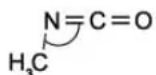
120°

C.



109°

D.

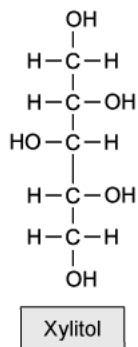


104°

[1 mark]

Question 6

Xylitol is an artificial sweetener used in toothpastes and chewing gum to improve their taste without impairing dental hygiene.



Which functional groups could be produced if Xylitol is oxidised under suitable conditions?

- 1 Alkene
- 2 Aldehyde
- 3 Carboxylic acid
- 4 Ketone

A. 1 only

B. 2 only

C. 2, 3 and 4

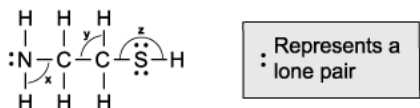
D. 2 and 4

[1 mark]

Question 7

A synthetic molecule called N-acetylcysteine is used as an antidote for patients who have ingested too many paracetamol tablets.

The diagram shows part of the molecule N-acetylcysteine and bond angles have been labelled.



Which is the correct size order of bond angles x, y, z from smallest to largest?

- A. $x \rightarrow y \rightarrow z$
- B. $y \rightarrow z \rightarrow x$
- C. $z \rightarrow y \rightarrow x$
- D. $z \rightarrow x \rightarrow y$

[1 mark]

Question 8

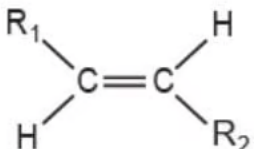
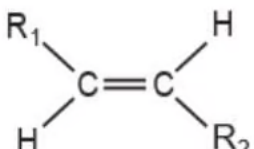
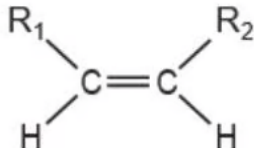
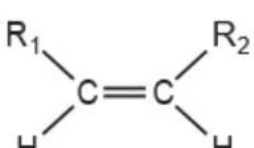
11-*cis* retinal, $C_{20}H_{28}O$, is found in the retina of animal eyes and is required for vision.

An 11-*cis* retinal molecule contains the following functional groups:

- 1 x aldehyde group
- 1 x cyclohexene ring
- 1 x aliphatic side chain

Within the side chain, there is a double bond between carbons 11 and 12.

Which letter is associated with the correct pair?

	Arrangement of atoms around the adjacent carbons 11 and 12	Total number of carbon-carbon double bonds
A		5
B		6
C		5
D		6

[1 mark]

Question 9

1-chloroethane, $\text{CH}_3\text{CH}_2\text{Cl}$ and NaOH (aq) are reagents in a reaction.

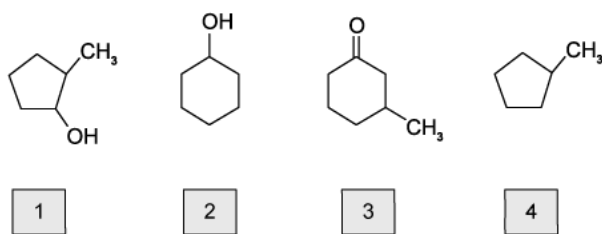
Which row correctly describes what type of reaction this would be?

- A. Nucleophilic addition
- B. Nucleophilic substitution
- C. Electrophilic addition
- D. Electrophilic substitution

[1 mark]

Question 10

Which of the compounds shown have only 1 chiral carbon atom?



- A. 2 and 4
- B. 4 only
- C. 1 and 3
- D. 3 only

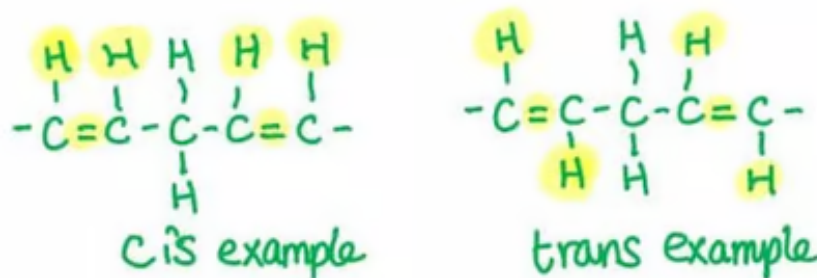
[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

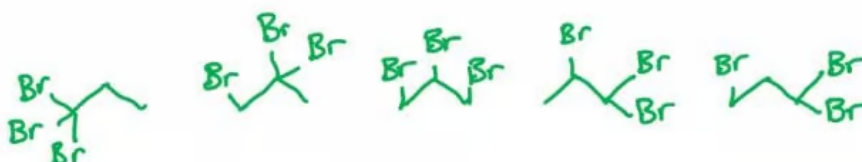
- Stereoisomers are compounds only different by the **spatial arrangement** of the atoms in the molecule.
- One type of stereoisomerism is **optical isomerism**.
 - These are non-superimposable mirror images of each other around a chiral carbon.
 - A chiral carbon is one with 4 different bonds attached.
 - No carbon atoms in this molecule are chiral.
- Another type of stereoisomerism is **geometric**.
 - This is shown below showing examples of **cis-trans isomers** of gamma-linolenic acid.
 - In **trans** isomers, the groups with the **largest M_r** are on either side of the double bond are on **different** sides of the double bond plane.
 - In **cis** isomers, the groups with the **largest M_r** on either side of the double bond are on the **same** side of the double bond plane.
- The number of stereoisomers from gamma-linolenic acid is 8 as the isomerism at each double bond can be as follows:
 1. cis, cis, cis
 2. cis, cis, trans
 3. cis, trans, trans
 4. cis, trans, cis
 5. trans, cis, cis
 6. trans, cis, trans
 7. trans, trans, cis
 8. trans, trans, trans



2

The correct answer is **B** because:

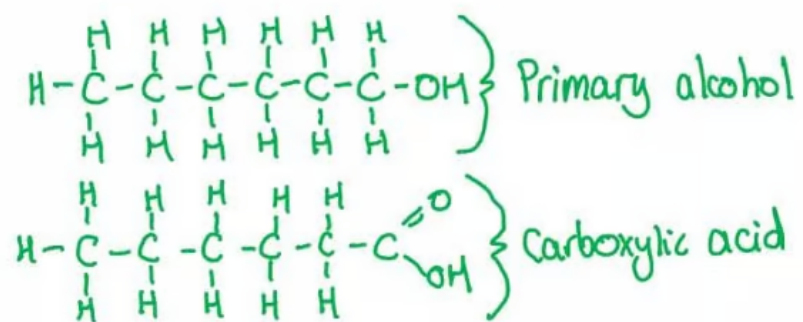
- Structural isomers have the atoms arranged in a completely different order.
- These can be chain isomers (branched chains), position isomers (groups moved on the carbon skeleton) and functional group isomerism (different functional groups and a member of a different homologous series).
- The isomers of C₃H₅Br₃ are shown below:



3

The correct answer is **C** because:

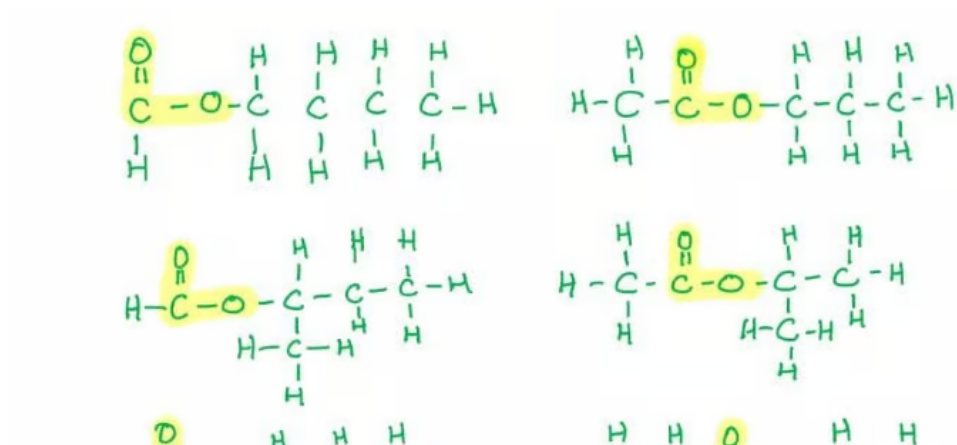
- C₆H₁₄O is an alcohol, and the functional group of the alcohol is OH.
- This has been oxidised to a carboxylic acid C₆H₁₂O₂.

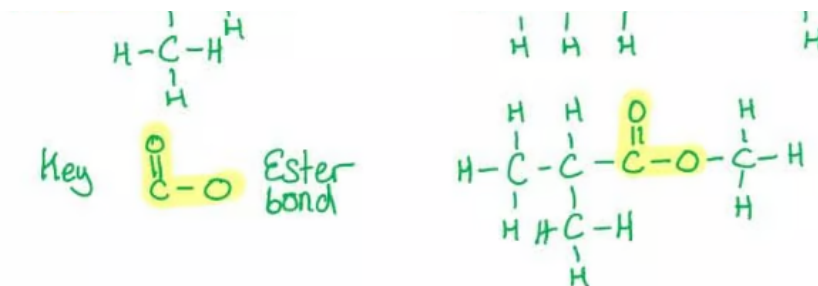


4

The correct answer is **C** because:

- Structural isomers have the atoms arranged in a completely different order.
- Isomeric esters are esters with a different location of the functional group.
- The possible isomers of $C_5H_{10}O_2$ are shown below:

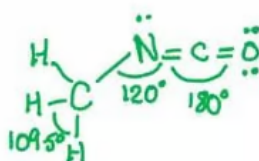




5

The correct answer is **B** because:

- Firstly, identify the number of **bond pairs** and **lone pairs** of electrons- in this case, nitrogen will have 1 lone pair of electrons.
- When working out bond angles, you then need to count the number of '**electron groups**', which are the areas of electron density around the central atom, in the case of nitrogen it has **3 electron groups**:
 1. A bonding pair of electrons with the methyl group
 2. A lone pair of electrons
 3. A double bond (2 bonding pairs of electrons) to the C=O carbon
- The **VESPR theory** states that electron pairs **repel** each other whether or not they are in bond pairs or lone pairs; therefore the electrons will spread as much as possible to **reduce repulsion**.
- Therefore, the bond angle is $(360 / 3 =) 120^\circ$.



Technically, the bond angle will be slightly **less than 120°** because the lone pair generates a greater degree of repulsion than bonding pairs – lone pairs repel by an extra 2.5° . This will cause the angle between the two bonding electron groups to decrease below 120° . However, the question asked for the 'approximate' angle, and 120° is the closest of the possible options.

6

The correct answer is **C** because:

- A primary alcohol has a hydroxyl group (-OH) attached to a saturated carbon atom that has only 1 other carbon atom attached to it.
- Secondary alcohol has a hydroxyl group (-OH) attached to a saturated carbon atom that has 2 other carbon atoms.
- Primary alcohols are oxidised to aldehydes and carboxylic acids.
- Secondary alcohols are oxidised to ketones.

7

The correct answer is **D** because:

- Y will be the largest because the carbon is the central atom with a **tetrahedral** shape. All the bond angles in a tetrahedral are **109.5°**
- In X the central atom is the nitrogen, and this has a lone pair so will have a **trigonal-pyramidal** shape with a bond angle of **107°**
- Z will be **bent / V-shaped**, due to the repulsion of the two lone pairs the bond angle decreases to **104.5°**

VESPR theory of electron repulsion:

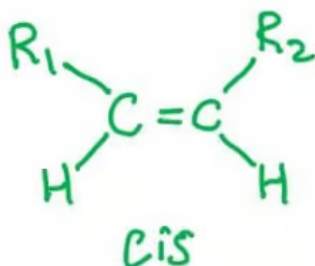
All electron pairs repel each other due to all being negatively charged, and therefore will space out around a central atom as far away as possible. The amount of repulsion given by lone pairs is **more** than by bonding pairs, so the amount of repulsion between electron pairs from highest to lowest is : lone pair – lone pair > lone pair – bond pair > bond pair – bond pair.

Hint: For every **lone pair** of electrons present around the central atom, the bond angle **reduces** by **2.5°** .

8

The correct answer is **C** because:

- The question states that it is a *cis* isomer, this means that the groups with the largest M_r (usually the functional groups or groups other than hydrogen) are the same side of the double bond as seen below.



- The ring structure is a hexene so has one double bond within the ring, there are 4 remaining carbon-carbon double bonds on the aliphatic chain.

A & B are incorrect as they both show *trans* isomers.

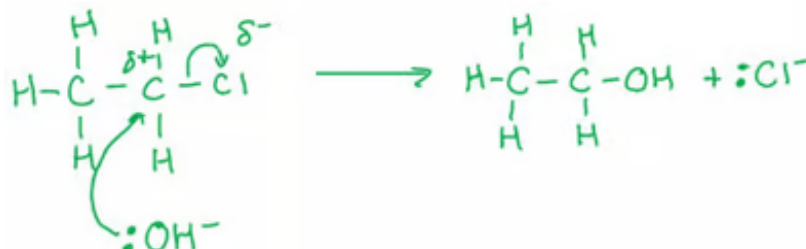
D is incorrect as this includes the double bond with the oxygen; the question states only carbon-carbon double bonds.

9

The correct answer is **B** because:

- When a halogenoalkane is heated under reflux with a solution of sodium or potassium hydroxide, the OH^- ion will replace the halogen making an alcohol.
- The hydroxide ion is a **nucleophile**; an **electron-pair donor**.
- The hydroxide ion is taking the place of the halogen in the compound, so this is a **substitution** reaction. $\Rightarrow$ Nucleophilic substitution. (The reaction is deemed electrophilic/nucleophilic dependent on how the **inorganic reagent** acts).
- The reaction can be summarised into:

- The hydroxide ion is a **nucleophile**; an **electron-pair donor**.
- The hydroxide ion is taking the place of the halogen in the compound, so this is a **substitution** reaction. $\Rightarrow$ Nucleophilic substitution. (The reaction is deemed electrophilic/nucleophilic dependent on how the **inorganic reagent** acts).
- The reaction can be summarised into:



10

The correct answer is **D** because:

- **Chiral** carbons are carbon atoms that are bonded to **4 different groups**.
- In cyclic molecules, it is important to look at whether the molecule is **symmetrical** on both sides of the potential chiral carbon.
- 1 and 3 are **not symmetrical** along a line from the chiral carbon and therefore are **chiral**.
- 2 and 4 are symmetrical along a line from the chiral carbon and therefore are not chiral.
- **3** is the only molecule with **1 chiral carbon**.
-

